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Magnetic properties of the Alnön Complex

JOHN D. A. PIPER

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The plutonic rocks, sövite ring dykes and alnöite dykes belonging to the (Cambrian) Alnön Complex have been analysed for palaeomagnetic study. Total NRM's residing in magnetite and pyrrhotite are locally very high in the plutonic facies and variable in the sövite dykes; all the alkaline phases have NRM's dominated by low coercivity components. Alternating field demagnetisation isolates a range of remanence directions, but a principal mean of $D = 107°$, $I = 2°$ ($\alpha_{95} = 9°$) in the bulk of the sövite and alnöite dykes is defined by both polarities and confirmed by thermal cleaning. A subsidiary direction of $\sim D = 350°$, $I = 75°$ ($\alpha_{95} = 11°$) is recognised predominantly in the plutonic rocks. The data contribute to the sparse Cambrian palaeomagnetic record from the Fennoscandian Shield and support earlier suggestions of large apparent polar wander motion in Cambrian times. □ *Palaeomagnetism, thermomagnetism, polar wandering, pole positions, dykes, plutonic rocks, sövite, alnöite, Alnön Complex, Fennoscandian Shield, Cambrian, Alnön, eastern Medelpad, east central Sweden, N6226 N6229 E1729 E1724.*

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Very little is at present understood about continental movements and the behaviour of the Earth's magnetic field in Cambrian times, and such data as are presently available suggest that large apparent polar wander (a.p.w.) movements characterised this period (Bucha 1965; McElhinny et al. 1974; Morris & Roy 1977). A thorough analysis of Cambrian rocks will be required to resolve these movements. The present paper reports a palaeomagnetic study of the Alnön Complex falling within this general objective. All radiometric ages quoted have been recalculated using the new decay constants recommended by IUGS (Steiger & Jäger 1977).

Age of the Alnön Complex

The Alnön Complex comprises a suite of alkaline plutonic rocks ranging from pyroxenite to ijolite–melteigite and nepheline syenite in composition. They are surrounded by a fenite aureole, and cut by sövite ring-dykes. The main intrusion at Alnön Island is accompanied by several satellite intrusions: Båräng (sövite), Sälskär (breccia), Söråker (melilite-rich rocks) and possibly Åvike Bay (sövite?). Various dyke rocks, including alkaline dykes, carbonatite dykes and alnöites, occur scattered in an area of about 1000 km². The most recent account on the geology of the Alnö Complex has been given by Peter Kresten (unpublished mimeographed report, Nordic Carbonatite Symposium, Alnö, May 1979; 67 pp.).

Von Eckermann & Wickman (1956) estimated a maximum (K–Ar) age of 571 Ma from a sövite ring dyke and Doig (1970) quoted a K–Ar age of 589 Ma from biotite. More recent estimates are marginally younger: Welin et al. (1972) dated phlogopite from alnöite breccia at 564 Ma. Kresten et al. (1977) dated phlogopite from a sövite ring dyke at 545 Ma and from an alnöite dyke of the Åvike Bay area (15 km ENE of the Alnön Complex) at 550 Ma (Wilson & Sundin 1979). Most recently Brueckner & Rex (1979) have determined a 553 ± 6 Ma Rb–Sr isochron from alkaline dykes surrounding the main complex.

Palaeomagnetic sampling and processing

Cores 2.5 cm in diameter were drilled at 32 localities (Fig. 1) in the plutonic rocks, sövites, and alnöite dykes, and oriented wherever possible by solar compass and otherwise by magnetic compass in conjunction with topographic sights. The bulk of these localities are on the northern half of Alnön Island but numbers 25–32 are in alnöite dykes 1–2 km north of Söråker. Magnetic moments were measured with a parastatic magnetometer system before and after progressive treatment in alternating fields; the peak

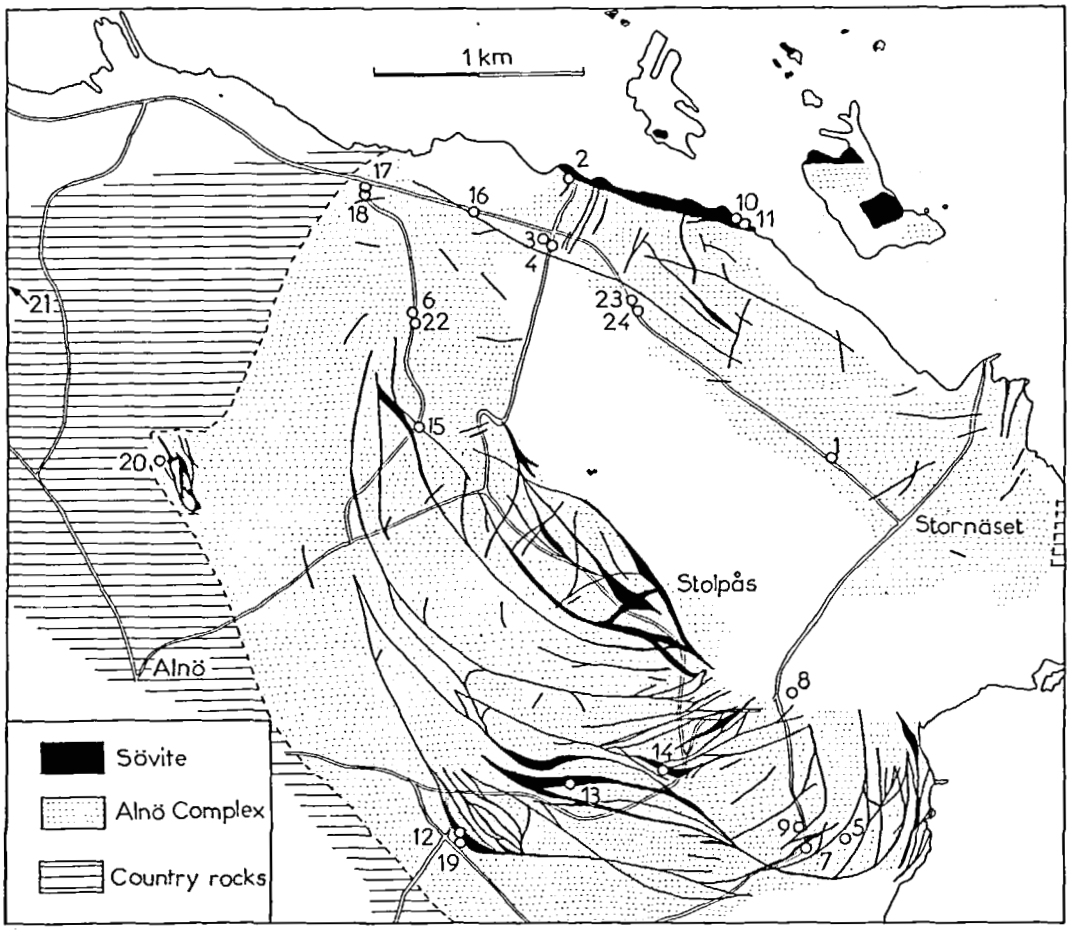


Fig. 1. Location of palaeomagnetic sampling sites in the Alnö Complex; the outcrop of the major geological units is based on Kresten (1979).

values of these fields were increased in steps of 5 or 10 mT (milli-Tesla) until moments became either too weak to measure or directional changes became erratic. Selected samples were also thermally cleaned at progressively-increasing temperatures; susceptibilities were determined with a standard bridge system and Curie points (saturation magnetisation, J_s, *versus* temperature) were determined with a modified Chevallier balance.

Palaeomagnetic results

From ground surveys Kresten (1976) found highest magnetic fields associated with the alkaline plutonic rocks; somewhat lower fields characterise high-grade fenite terrains intruded by alkaline carbonatite intrusions, while low-grade fenites have fields only marginally above the Svecofennian country rock. The laboratory determinations define a comparable pattern: total NRM's and susceptibilities from the plutonic rocks (Table 1) are locally very high; determinations from the sövites exhibit extreme variation apparently reflecting the sporadic distribution of magnetite in these rocks, while determinations from the alnöite dykes are more uniform.

Thermomagnetic determinations yield Curie points of 510–580°C attributable to magnetite (pure magnetite $T_c \sim 575$°C) at most sites while several also exhibit a lower Curie point at ca. 300°C (Fig. 2). These two Curie points are unlikely to result from two phases of titanomagnetite (Ti-rich and Ti-poor) both because of the consistency of the lower point, and because Curie points lower than 500°C are uncommon in older rocks presumably due to gradual unmixing

of titaniferous phases with time. The lower Curie point appears to represent a separate phase and the possible contenders are the sulphide pyrrhotite (Curie point at ca. 310°C but with some ferrimagnetic properties to 560°C) and maghemite a peripheral and fracture-related alteration phase after magnetite (this mineral alters to hematite with low saturation remanence during heating in thermomagnetic experiments).

Hematite has not been positively identified by this method at any sites and is known to be very rare (Peter Kresten, Uppsala, personal communication 1980). Alternating field (a.f.) demagnetisation studies show that the bulk of the total NRM is of low or very low coercivity (Fig. 3) confirming the presence of the carriers inferred from thermomagnetic analysis (Nagata 1967), and most samples have only 1–10% of their initial remanence remaining after treatment in median destructive fields of 10–50 mT. Low coercivities accompanied by poor directional stability in many samples imply a predominance of multidomain carriers (Nagata 1967; Lowrie & Fuller 1971); probably for this reason, thermal demagnetisation studies achieved only limited success. In many samples erratic changes in moment and susceptibility suggest that the heating procedure (in air) tends to promote chemical changes, and only a few of the more successful examples are illustrated in Fig. 3.

Progressive a.f. cleaning of the plutonic rocks identified several directions (Fig. 3). At site 1 a shallow northerly remanence is also retained by thermal cleaning to 550°C but a comparable remanence at site 4 is removed above 10 mT to converge to a steep positive direction comparable to sites 2, 4, 7 and 8. Only a few localities show univectional remanence and at site 8 for example a minority of samples exhibit a remanence ($\sim D = 98$, $I = +18$) comparable to sites 5 and 9, whilst the predominant (steep) remanence has higher coercivity.

Like the plutonic phases, strongly-magnetised sövites have very low coercivity remanence and where end points are identified they reside in less than 1% of the total NRM. The majority of samples migrate to easterly or westerly shallow inclination directions (Fig. 3) and at sites 12 and 15 different samples migrate to nearly antiparallel directions; a comparable minority direction at sites 14 and 15 contrasts with steeper directions of variable declination in the remaining samples.

In the alnöite dykes the majority of remanence vectors also move to shallow easterly and westerly directions with a.f. cleaning defined by a few percent of initial remanence; the stability

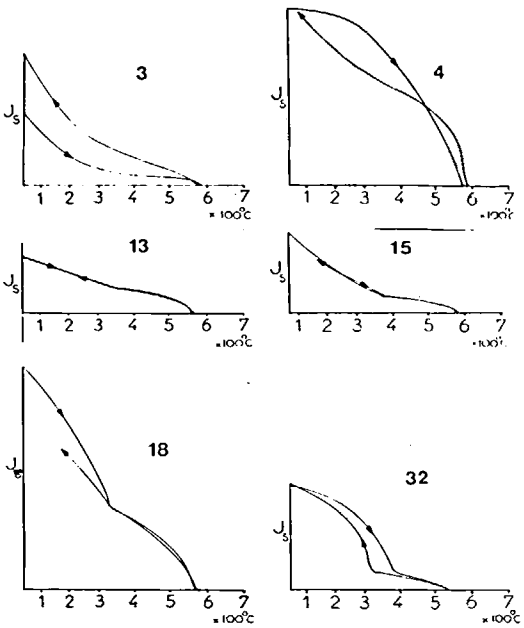


Fig. 2. Some typical thermomagnetic determinations (saturation magnetisation, J_s, *versus* temperature) from rocks of the Alnö Complex; the J_s axis is arbitrary.

of this direction has also been confirmed by thermal demagnetisation at some sites (Fig. 3). At sites 16, 18 and 20 different samples move to antiparallel directions and since both pyrrhotite and magnetite are identified in the thermomagnetic spectra (Table 1) it is possible that these components reside in different phases. Subsidiary directions reside in weak components at sites 18, 19 and 25 and their relationship to the predominant E–W remanence is not clear. Sites 25–32 from a suite of thin alnöite dykes north of Söråker include a shallow easterly component common to sites 25 to 28, but at site 27 this is removed at low fields to recover a direction ($D = 340°$, $I = 80°$) which is present as the only component in some other samples. A comparable but antiparallel remanence is present as a stable component in two samples of site 32 (Fig. 3) and identified by thermal cleaning at site 30. The shallow northerly component present at site 25, is observed at some other sites but is removed by low a.f. fields in these latter cases.

The distribution of sample directions is reflected in the overall distribution of site mean directions (Fig. 4). Steep directions isolated by a.f. cleaning at several sites in the plutonic facies (2, 4, 7, 8) and two alnöite dykes (22, 27) yield a group mean ('B') of $D = 348°$, $I = 76°$ ($\alpha_{95} = 11°$)

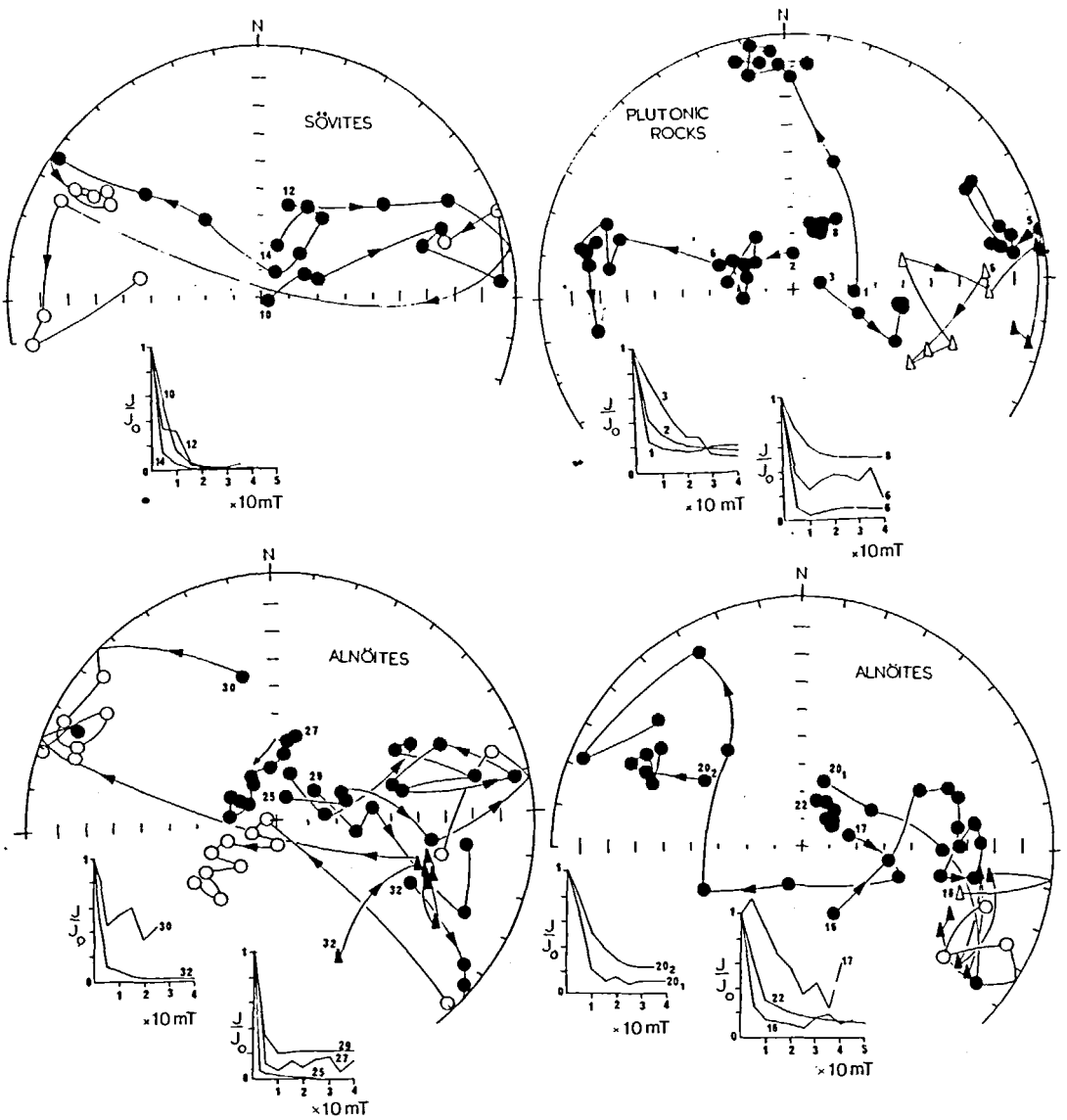


Fig. 3. Behaviour of some samples from rock groups belonging to the Alnö Complex with progressive a.f. (circles) and thermal (triangles) treatment. Stereographic projections; upper hemisphere plots are open symbols and lower hemisphere plots are closed symbols. The decay of initial remanence is expressed as a fraction of the initial moment J_o.

and pole position at 299°E, 85°N; a mean calculation based on all (42, 3 reversed) samples yielding steep directions gives a mean $D = 7°$, $I = 81°$ ($\alpha_{95} = 5°$), which is not significantly different, and a pole at 29°E, 79°N. Unfortunately, the stability of this direction was only locally confirmed by thermal cleaning. The clearest picture to emerge from the a.f. and/or thermal cleaning is the definition of shallow directions with easterly and westerly declinations (Fig. 4). This 'A' direction ($D = 107°$ $I = 2°$, $\alpha_{95} = 9°$) includes the majority of the sövite and alnöite dykes; no mean is calculated for the minority of sites exhibiting northerly and southerly shallow directions (Fig. 4) because these are not collectively well-grouped and may not represent a coherent magnetisation. Both these 'A' and 'B' directions are widely different from Ordovician and later magnetic

Table 1. Palaeomagnetic results, Alnö Complex (17.5°E, 62.5°N). *D* and *I* are the mean declination and inclinations calculated from *n* samples (total for the site = *N*) and *R* is the resultant vector; *k* is the precision parameter and α_{95} the radius of the 95 percent confidence circle about the mean direction. *Dp* and *dm* are the radii of the oval of confidence about the pole position in the colatitude direction and at right angles to it respectively.

Site No.	J_o ($\times 10^{-5}$ A·m²/kg)	N/n	D	I	R	Mass susceptibility ($\times 10^{-5}$ SI units)	Curie point(s) (°C)
Phase 1. Plutonic rocks							
1	7.550–0.415	7/6	0	15	5.78	8.463–1.872	510
2	1.718–0.276	7/4	309	78	3.92	1.195–0.020	570
3	0.813–0.009	6/5[a]	104	53	4.93	0.870–0.064	300, 570
4	0.598–0.019	6/6	39	77	5.84	0.206–0.077	285, 570
5	0.299–0.015	6/4	87	–9	3.92	0.615–0.052	575
6	1.947–0.073	7/5	113	–22	4.82	1.063–0.091	310–565
7	0.232–0.010	5/5	312	60	4.86	0.114–0.070	300, 555
8	0.389–0.015	6/5	5	67	4.64	3.860–0.031	555
9	461.789–1.231	7/7	123	17	6.89	7.725–0.156	555
Phase 2. Sövite centre and dykes							
10	6.774–0.004	6/4	84	–6	3.94	7.204–0.013	580
11	12.175–0.003	7/5	297	–8	4.39	1.096–0.046	575
12	0.052–0.001	8/5	299	3	4.60	0.052–0.039	550
13	6.963–0.115	7/7	109	–20	6.31	2.510–0.049	330, 560
14	16.522–0.367	7/2	302	–16	1.99	3.860–0.020	550
15	0.229–0.005	7/4	284	9	3.77	0.095–0.021	330, 570
Phase 3. Alnöite dykes							
16	2.620–0.009	6/6	104	11	5.56	5.475–0.073	310, 580
17	0.313–0.002	6/6	287	24	5.57	0.083–0.045	570
18	1.176–0.003	7/6	41	27	5.79	1.313–0.049	325, 570
19	0.064–0.007	5/3	14	26	2.91	0.065–0.031	320, 565
20	1.391–0.046	6/6	120	–15	5.84	1.929–0.081	560
21	2.995–0.002	7/7	123	2	6.23	1.816–0.033	565
22	1.089–0.218	6/6	42	77	5.98	0.024–0.014	565
23	3.695–1.294	4/4	264	7	3.67	5.581–2.619	570
24	4.390–1.847	9/9	84	9	8.61	5.719–3.950	560
25	7.157–2.256	(I) 6/3 (II) 6/2	90 28	24 28	2.96 1.83	–	575
26	25.950–0.548	7/7[a]	104	40	6.59	–	565
27	0.298–0.019	9	338	80	3.94	–	570
28	12.245–4.170	5	103	24	4.69		495, 550
29	9.175–0.112	4				–	580
30	0.164–0.074	7/6	306	–12	5.77	–	580
31	1.066–0.026	5/4	88	17	3.40	–	580
32	5.952–4.930	4/4	119	20	3.90	–	320, 530

Group mean directions and pole positions

	D	I	N/R	k	α_{95}	*Palaeomagnetic pole* °E	°N	(*dp*/*dm*)
'A' directions	107.2	1.6	21/19.41	13	9.3	92.0	–7.6	(5/9)
'B' directions	348.4	75.8	6/5.87	37	11.1	299.4	84.7	(18/20)

[a] Excluded from group mean calculations.

field directions for Europe (Irving et al. 1976). Since they reside in components of magnetite, and a general blanket remagnetisation due to hematite has not been identified, they probably relate to the interval of initial cooling of the Alnö Complex. The 'B' direction predominates in the plutonic facies and may be a later-cooling magnetisation, but it has not yet proved possible to confirm its stability by thermal demagnetisation. Accordingly, it may be of composite origin and no further interpretation is attempted here.

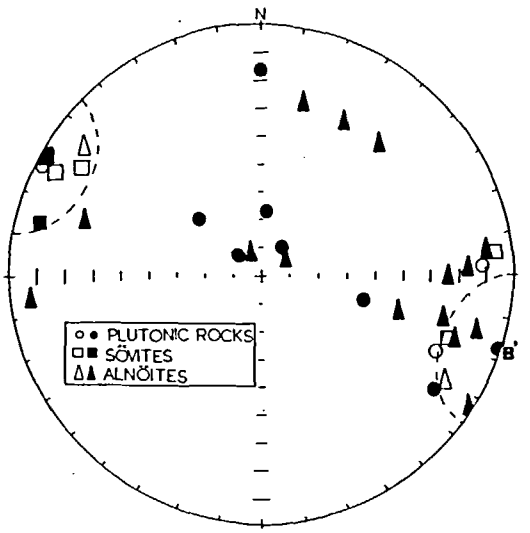


Fig. 4. Site mean directions of magnetisation from rock groups belonging to the Alnö Complex.

Palaeomagnetic poles

There are currently very few Cambrian palaeomagnetic results from the tectonic block incorporating the Fennoscandian Shield with which to compare these data, and Prasad & Sharma (1978) note the wide differences between the pole determinations. The analogous Fen Complex in southern Norway has produced a range of age determinations between 613 and 522 Ma but mostly lying between 575 and 599 Ma (Broch 1964; Priem et al. 1968). Highly-stable hematite-bearing carbonatite from this complex yields an a.f. mean $D = 205°$, $I = -56°$ (Poorter 1972), although several other (probably hematite) components with shallow inclinations are also suggested by thermal demagnetisation of a variety of Fen rocks (Storetvedt 1973, Figs. 3–4).

The wide difference in pole positions 1 to 4 from the Fennoscandian Shield (Table 2), and their divergence from Lower Ordovician data, merely emphasises the conclusions of previous workers (eg., McElhinny et al. 1974; Prasad & Sharma 1978) that large a.p.w. movements took place in Cambrian times, and it is premature to suggest an a.p.w. path for this period.

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Table 2. Cambrian palaeomagnetic poles from the Fennoscandian Shield. 1. Poorter (1972); 3. Prasad & Sharma (1978); 2 and 4. This study.

Rock unit	Age (Ma)	*D/I*	Palaeomagnetic pole °E	°N	*dp/dm*
1. Fen Rødberg	613–522	205/–56	142	63	3/4
2. Alnö Complex 'A'	553–537	107/1	272	8	5/9
3. Nexø Sandstone	Late Precambrian–Early Cambrian	226/–30	134	38	7/11
4. Alnö Complex 'B'	–	348/76	348	76	19/20
Mean Early Ordovician pole for the Russian Platform: 150°E 39°N					

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